Product Rule

Exam code: 9MA0

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Updated on 16 February 2026

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Product rule

What is the product rule?

- The product rule is a formula that allows you to differentiate a **product of two functions**
- If $y = u \times v$ where u and v are functions of x then the product rule is:

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

- In **function notation**, if $f(x) = g(x) \times h(x)$ then the product rule can be written as:

$$f'(x) = g(x)h'(x) + h(x)g'(x)$$

- The easiest way to remember the product rule is, for $y = u \times v$ where u and v are functions of x:

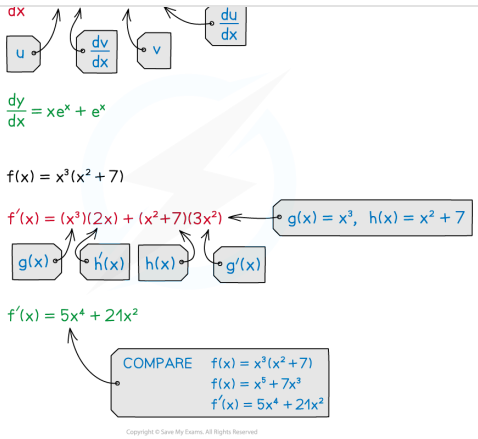
$$y' = uv' + vu'$$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Examiner Tips and Tricks

- The product rule formulae are NOT in the formulae booklet – you need to know them.
- Don't confuse the **product** of two functions with a **composite function:**
  - The **product** of two functions is two functions **multiplied** together
  - A **composite** function is a **function of a function**

e.g. $f(x) = 3x^2$ $\quad g(x) = x - 3$

$f(x)g(x) = 3x^2(x - 3) = 3x^3 - 9x^2$ PRODUCT

$fg(x) = f(g(x)) = 3(x - 3)^2 = 3x^2 - 18x + 27$ COMPOSITE FUNCTION

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- To differentiate composite functions you need to use the **chain rule**

Worked Example

a) $(5\sin x - 7\cos x)e^{3x+2}$

b) $5x\ln 5x$

a) $u = 5\sin x - 7\cos x$ $\quad$ $v = e^{3x+2}$

$\frac{du}{dx} = 5\cos x + 7\sin x$ $\quad$ $\frac{dv}{dx} = 3e^{3x+2}$

DON'T FORGET MINUS SIGN WHEN YOU DIFFERENTIATE cos

USE CHAIN RULE

$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

$\frac{dy}{dx} = (5\sin x - 7\cos x)(3e^{3x+2}) + (e^{3x+2})(5\cos x + 7\sin x)$

$\frac{dy}{dx} = (22\sin x - 16\cos x)e^{3x+2}$

b) $u = 5x$ $\quad$ $v = \ln 5x$

$\frac{du}{dx} = 5$ $\quad$ $\frac{dv}{dx} = \frac{1}{x}$

BE CAREFUL WITH THIS ONE! BECAUSE OF THE CHAIN RULE THE DERIVATIVE OF $\ln(ax)$ IS $\frac{1}{x}$ FOR **ANY** CONSTANT a

$\frac{dy}{dx} = (5x)\left(\frac{1}{x}\right) + (\ln 5x)(5)$

$\frac{dy}{dx} = 5 + 5\ln 5x$

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